

1 Tangent Space

• Let S be a regular parametric surface that is given by $S : \mathbf{r} = \mathbf{r}(u, v)$. Then a smooth curve C on S can be given by $C : \mathbf{r}_0(t) = \mathbf{r}(u(t), v(t))$ with $u(t), v(t) \in C^\infty(\mathbb{R}; \mathbb{R})$.

Definition 1.(Tangent Vectors)

Let $S \hookrightarrow \mathbb{R}^3$ be a regular surface and $p \in S$. A vector v is said to be a **tangent vector** to S at p if it is a tangent vector $\mathbf{r}'(t_0)$ of some smooth curve $C : \mathbf{r} = \mathbf{r}(t)$ on S with $\mathbf{r}(t_0) = p$ and $\mathbf{r}'(t_0) = v$.

Proposition 1. Let $S : \mathbf{r} = \mathbf{r}(u, v)$ be a regular parametric surface and $p \triangleq \mathbf{r}(u_0, v_0) \in S$, then for any tangent vector v of S at $p \in S$, we have

$$v \in \text{Span}_{\mathbb{R}}\{\partial_u \mathbf{r}|_{(u_0, v_0)}, \partial_v \mathbf{r}|_{(u_0, v_0)}\}.$$

Conversely, if a vector v satisfies

$$v \in \text{Span}_{\mathbb{R}}\{\partial_u \mathbf{r}|_{(u_0, v_0)}, \partial_v \mathbf{r}|_{(u_0, v_0)}\},$$

then the vector v is a tangent vector of S at $p \in S$.

Proof: If v is a tangent vector of S at $p \triangleq \mathbf{r}(u_0, v_0) \in S$, then there is a curve $C : \mathbf{r}(t) = \mathbf{r}(u(t), v(t))$ on S such that $\mathbf{r}(t_0) = \mathbf{r}(u(t_0), v(t_0)) = \mathbf{r}(u_0, v_0) = p$ and $v = \mathbf{r}'(t_0)$, thus by the chain rule we get

$$v = \left. \frac{d\mathbf{r}(t)}{dt} \right|_{t=t_0} = \left. \frac{\partial \mathbf{r}}{\partial u} \right|_{(u_0, v_0)} \cdot \left. \frac{du(t)}{dt} \right|_{t=t_0} + \left. \frac{\partial \mathbf{r}}{\partial v} \right|_{(u_0, v_0)} \cdot \left. \frac{dv(t)}{dt} \right|_{t=t_0},$$

i.e.,

$$v = u'(t_0) \cdot \partial_u \mathbf{r}|_{(u_0, v_0)} + v'(t_0) \cdot \partial_v \mathbf{r}|_{(u_0, v_0)},$$

which means $v \in \text{Span}_{\mathbb{R}}\{\partial_u \mathbf{r}|_{(u_0, v_0)}, \partial_v \mathbf{r}|_{(u_0, v_0)}\}$;

Conversely, if $v \in \text{Span}_{\mathbb{R}}\{\partial_u \mathbf{r}|_{(u_0, v_0)}, \partial_v \mathbf{r}|_{(u_0, v_0)}\}$, then we may write

$$v \triangleq a \cdot \partial_u \mathbf{r}|_{(u_0, v_0)} + b \cdot \partial_v \mathbf{r}|_{(u_0, v_0)} \text{ with } a, b \in \mathbb{R}.$$

Then by defining the curve $\mathbf{r}(t) := \mathbf{r}(u_0 + at, v_0 + bt)$ on S , we see that $\mathbf{r}'(t_0) = a \cdot \partial_u \mathbf{r}|_{(u_0, v_0)} + b \cdot \partial_v \mathbf{r}|_{(u_0, v_0)} = v$. ■

Definition 2.(Tangent Space)

Let $S : \mathbf{r} = \mathbf{r}(u, v)$ be a regular parametric surface and $p \triangleq \mathbf{r}(u_0, v_0) \in S$, the **tangent space** of S at $p \in S$ is the vector space consists of all tangent vectors at p , which is denoted by $T_p S$.

Remark: By proposition 1, we immediately obtain that

$$T_p S = \text{Span}_{\mathbb{R}}\{\partial_u \mathbf{r}|_{(u_0, v_0)}, \partial_v \mathbf{r}|_{(u_0, v_0)}\}, \quad \dim_{\mathbb{R}} T_p S = 2,$$

whose basis can be given by $\{\partial_u \mathbf{r}|_{(u_0, v_0)}, \partial_v \mathbf{r}|_{(u_0, v_0)}\}$ since $\partial_u \mathbf{r} \times \partial_v \mathbf{r} \neq \mathbf{0}$ because S is regular. And the parametric equation for the tangent space $T_p S$, which is also a 2-dimensional plane is given by

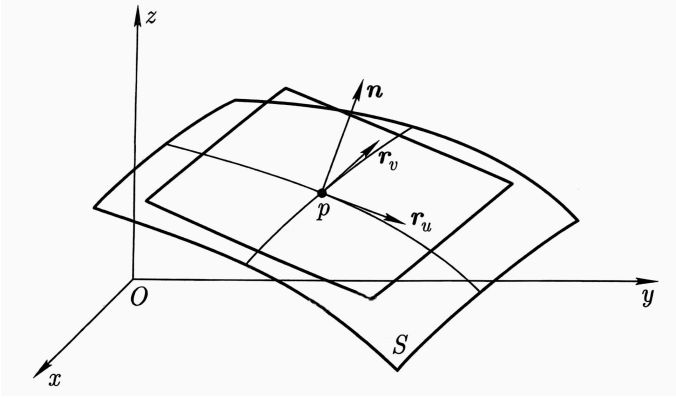
$$T_p S : \mathbf{X}(t, s) = \mathbf{r}(u_0, v_0) + t \cdot \partial_u \mathbf{r}|_{(u_0, v_0)} + s \cdot \partial_v \mathbf{r}|_{(u_0, v_0)}, \quad (s, t) \in \mathbb{R} \times \mathbb{R},$$

where we denote $p = \mathbf{r}(u_0, v_0) \in S$. It's clear that a normal vector for the plane $T_p S$, which is denoted as $\mathbf{n}_{T_p S}$, can be given by

$$\mathbf{n}_{T_p S} = \frac{\partial_u \mathbf{r}|_{(u_0, v_0)} \times \partial_v \mathbf{r}|_{(u_0, v_0)}}{\left| \partial_u \mathbf{r}|_{(u_0, v_0)} \times \partial_v \mathbf{r}|_{(u_0, v_0)} \right|}$$

the normal line $l_{\mathbf{n}_{T_p S}}$ is

$$l_{\mathbf{n}_{T_p S}} : \mathbf{X}(t) = \mathbf{r}(u_0, v_0) + t \cdot \frac{\partial_u \mathbf{r}|_{(u_0, v_0)} \times \partial_v \mathbf{r}|_{(u_0, v_0)}}{\left| \partial_u \mathbf{r}|_{(u_0, v_0)} \times \partial_v \mathbf{r}|_{(u_0, v_0)} \right|}, \quad t \in \mathbb{R}.$$



Remark: For a regular parametric surface $S : \mathbf{r} = \mathbf{r}(u, v)$, we get a *natural frame* $\{p = \mathbf{r}(u, v); \partial_u \mathbf{r}, \partial_v \mathbf{r}, \mathbf{n}_{T_p S}\}$ on every point $p = \mathbf{r}(u, v) \in S$.

Proposition 2. Let $f(x, y, z) \in C^\infty(\mathbb{R}^3; \mathbb{R})$ and $\nabla f := \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \neq \mathbf{0}$ at any point, then the set

$$f^{-1}(c) := \{(x, y, z) \in \mathbb{R}^3 \mid f(x, y, z) \equiv c\}$$

is a regular surface, where $c \in \mathbb{R}$ is a constant.

Proof: Let $p(x_0, y_0, z_0) \in f^{-1}(c)$, since $\nabla f|_p \neq \mathbf{0}$, w.l.o.g. we may assume that $\frac{\partial f}{\partial z}|_p \neq 0$. Hence by the implicit function theorem, we see that there is a neighborhood D of (x_0, y_0) and a function $z = z(x, y)$ defined on D such that

$$f(x, y, z(x, y)) \equiv c, \quad \forall (x, y) \in D.$$

Thus the parametric equation for the surface $f^{-1}(c)$ is

$$f^{-1}(c) : \mathbf{r} = \mathbf{r}(x, y) = (x, y, z(x, y)),$$

thus we get $\partial_x \mathbf{r} = (1, 0, \partial_x z)$, $\partial_y \mathbf{r} = (0, 1, \partial_y z)$ and hence $\partial_x \mathbf{r} \times \partial_y \mathbf{r} \neq \mathbf{0}$. ■

Proposition 3. Let $f(x, y, z) \in C^\infty(\mathbb{R}^3; \mathbb{R})$ and $\nabla f := \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \neq \mathbf{0}$ at any point, then the vector field ∇f is a normal vector field for the surface $S = f^{-1}(c)$.

Proof: Let $C : \mathbf{X}(t) = (x(t), y(t), z(t))$ be an arbitrary curve on $S = f^{-1}(c)$, which means

$$f(x(t), y(t), z(t)) \equiv c,$$

so we have

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx(t)}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy(t)}{dt} + \frac{\partial f}{\partial z} \cdot \frac{dz(t)}{dt} \equiv 0,$$

i.e.,

$$\nabla f \cdot \mathbf{X}'(t) \equiv 0,$$

by the arbitrariness of $\mathbf{X}(t)$ and hence $\mathbf{X}'(t) \in T_{\mathbf{x}(t)}S$, so we see that ∇f is a normal vector field of $S = f^{-1}(c)$. ■

• Now, we may consider the geometric meaning of $d\mathbf{r}(u, v)$ for a regular parametric surface $S : \mathbf{r} = \mathbf{r}(u, v)$. By the calculus of multi-variables, we get

$$d\mathbf{r}(u, v) = \partial_u \mathbf{r}(u, v) du + \partial_v \mathbf{r}(u, v) dv,$$

since

$$\mathbf{r}(u + \Delta u, v + \Delta v) - \mathbf{r}(u, v) = \partial_u \mathbf{r}(u, v) \Delta u + \partial_v \mathbf{r}(u, v) \Delta v + O\left(\sqrt{(\Delta u)^2 + (\Delta v)^2}\right),$$

so for a small displacement (du, dv) in the parameter domain, the corresponding displacement on the surface is approximated to first order by

$$\mathbf{r}(u + du, v + dv) - \mathbf{r}(u, v) \approx d\mathbf{r}(u, v) = \partial_u \mathbf{r} \cdot du + \partial_v \mathbf{r} \cdot dv.$$

Thus, $d\mathbf{r}$ is the **linear approximation** of the change in position on the surface and hence we see $d\mathbf{r}(u, v) \in T_{\mathbf{r}(u, v)}S$.